



GROUP 2

FINAL REPORT

A presentation about what our team has been doing on data science projects over the past period

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| 3 | Defining The Problem | 7 | Model Modeling |
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ABOUT US



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HOW WE COLLABORATE ?

Plan & Schedule





Nhập Môn Khoa Học Dữ Liệu _ 22KHDL1

Topic: Medicine Analysis

Health and wellness are important global topics. Especially in Vietnam, where users care about their needs.

Understanding the importance of providing accurate and transparent drug information, our team decided to collect data from Long Chau Pharmacy, one of Vietnam's most reputable and popular pharmacy chains. Through collecting and analyzing data from this pharmacy's website, we hope to help users easily look up and choose medicinal products that suit their needs, while ensuring safety, complete and reasonable price. The data we collect is taken from the source: nhathuoclongchau.com.vn

In business field, these data can provide valuable insights into market trends, pricing strategies, and consumer preferences, allowing them to optimize offerings, enhance their customer satisfaction, and stay competitive in the pharmaceutical industry.



Discussion

- 🔑 Topic selection
- ❓ Meaningful Questions
- 📄 Script For Progress Report
- ❓ Main Question
- 📄 Chọn thuộc tính cho model



 Add cover  Add comment

Workspace

URLs:

- Github: 🤖 [Introduction-to-Data-Science-22KHDL1](#)
- Notion: 📝 [Nhập Môn Khoa Học Dữ Liệu _ 22KHDL1](#)
- Drive: [IntroToDS_22KHDL1](#)
- Data Resource:

Plan			
Table	Timeline		
Công việc	Thời gian hoàn thành	👤 people	🔄 Status
Identify topic	October 16, 2024 → October 18, 2024	👤 Lý Liên Hoa 👤 Hưng Hoàng Duy 👤 Hoàng Đình 👤 Linh Lê Thị Ngọc	🟢 Done
Develop implementation plan	October 19, 2024	👤 Lý Liên Hoa 👤 Hoàng Đình 👤 Hưng Hoàng Duy 👤 Linh Lê Thị Ngọc	🟢 Done
Write the proposal	October 20, 2024 → October 21, 2024	👤 Hưng Hoàng Duy 👤 Lý Liên Hoa 👤 Hoàng Đình 👤 Linh Lê Thị Ngọc	🟢 Done
Data science process	October 21, 2024 → December 25, 2024	👤 Hưng Hoàng Duy 👤 Hoàng Đình 👤 Linh Lê Thị Ngọc 👤 Lý Liên Hoa	🟢 Done
Presentaion	December 25, 2024 → December 28, 2024	👤 Lý Liên Hoa 👤 Linh Lê Thị Ngọc	🟢 Done
Summary	December 25, 2024 → December 28, 2024	👤 Lý Liên Hoa 👤 Hưng Hoàng Duy 👤 Hoàng Đình	🟢 Done

Data science process

📅

Thời gian hoàn thành

📅

October 21, 2024 → December 25, 2024

🔄

Status

👤

people

+

Add a property

💬

Comments

🗨

Add a comment...

📄

Table

📅

Timeline

Task details

As Task	Estimated time	👤 Person	🔄 Status
Data collection	October 21, 2024 → October 31, 2024	Hung Hoang Duy Hoang Dinh	🟢 Done
Preprocess and clean data	October 31, 2024 → November 15, 2024	Hung Hoang Duy Hoang Dinh	🟢 Done
Data analysis and exploration	November 16, 2024 → November 30, 2024	Lý Liên Hoa Linh Lê Thị Ngọc Hung Hoang Duy Hoang Dinh	🟢 Done
Build and evaluate models	December 1, 2024 → December 24, 2024	Lý Liên Hoa Linh Lê Thị Ngọc Hung Hoang Duy Hoang Dinh	🟢 Done



HOW WE COLLABORATE ?

Version Control & Interact



Introduction-to-Data-Science-22KHDL1 Private Unwatch 1

main 10 Branches 0 Tags Add file <> Code

Switch branches/tags

Branches Tags

- ✓ main default
- DATA-MODELING
- DuyHung
- Explore-&-Analyze-Data
- FINAL
- HoangDinh
- Proposal-&-Collect-data-phase
- Scrape_Data_LongChau
- data_modeling
- progress_with_TA

[View all branches](#)

f8cd523 · 3 weeks ago	5 Commits
Initial commit	2 months ago
Push 2 file csv	3 weeks ago
Push 2 file csv	3 weeks ago
Basic frame for ipynb file	3 weeks ago

[a-Science-22KHDL1](#)

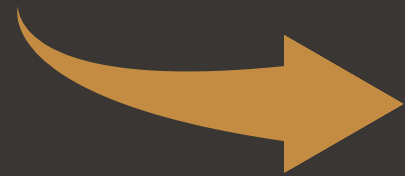
ence course



DEFINING THE PROBLEM

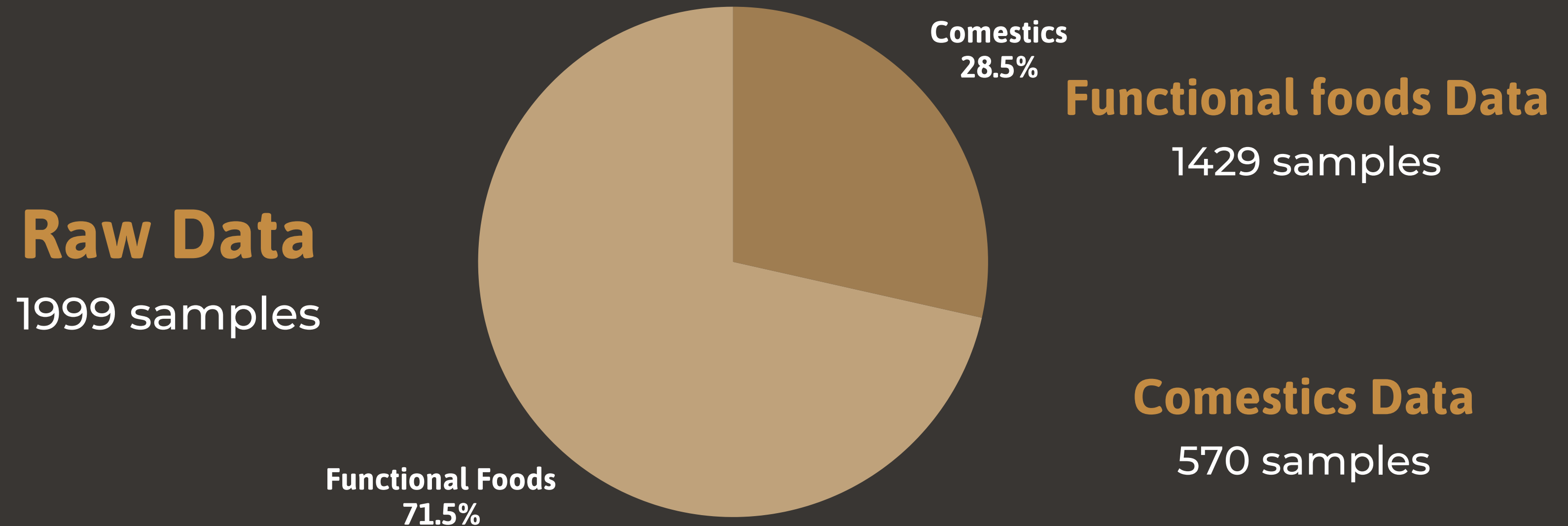


Customer needs



Based on Long Chau's supply, what factors affect customers demand?

DATA COLLECTION & PREPARATION





DATA PREPROCESSING

Meaning Of Rows & Columns

1	Product Name	Category	Dosage Form	Price	Trademark	Brand Origin	Country	Rating
2	Gel chấm mụn, mờ thâm Decumar Advance THC 20g dưỡng ẩm da, da mềm, mịn	Chăm sóc cơ thể	Gel	105000.0	DECUMAR	Việt Nam	Việt Nam	5.0
3	Dung dịch vệ sinh vùng kín Bimunica 250ml dành cho bé gái từ 0 tháng tuổi	Chăm sóc cơ thể		230000.0		Hoa Kỳ	Liên Bang Nga	5.0
4	Kem giảm thâm vùng nách, mông, bikini Neothera Armpit Cream La Beaute 35ml dưỡng trắng da	Chăm sóc cơ thể		139000.0	La Beauty	Việt Nam	Việt Nam	5.0
5	Gel rửa mặt SVR Sebiaclear Gel Moussant 200ml làm sạch không chứa xà phòng và loại bỏ tế bào da chết	Chăm sóc cơ thể		390000.0	SVR	Pháp	Pháp	
6	Bọt vệ sinh nam giới Sumely Men's Sanitary Foam giúp làm sạch, khử và ngăn chặn mùi vùng kín (100ml)	Lăn khử mùi, xịt khử mùi	Dạng bọt	96000.0		Việt Nam	Việt Nam	5.0
7	Sữa rửa mặt On: The Body Rice Therapy Heartleaf Acne Cleanser làm sạch sâu không gây khô da (150ml)	Sữa rửa mặt (Kem, gel, sữa)		132000.0		Hàn Quốc	Hàn Quốc	5.0
8	Nước tẩy trang JMsolution Water Luminous S.O.S Ringer Cleansing Water giúp làm sạch bụi bẩn, dầu thừa và lớp trang điểm (500ml)	Nước tẩy trang, dầu tẩy trang		199000.0		Hàn Quốc	Hàn Quốc	5.0
9	Gel Rosette Gommage Clear Peel tẩy tế bào chết (180g)	Tẩy tế bào chết	Gel	185000.0	Rosette	Nhật Bản	Nhật Bản	5.0
10	Gel làm mờ sẹo Fixderma Scars do mụn, bong, rạn da, vết thươngI (7ml)	Trị sẹo, mờ vết thâm		132800.0	Fixderma	Mỹ	Ấn Độ	5.0
11	Gội phủ bạc Healthy and lively H4 Lavox màu nâu cà phê (10 gói x 30ml)	Dầu gội đầu xả	Dung dịch	22400.0	LAVOX		Việt Nam	5.0

First 10 rows in origin data



DATA PREPROCESSING

Handling Missing Data

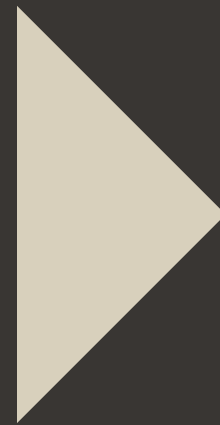
- Check percent of NULL values of each column except rating
- If the percent under or equal to 75%  Skip
- If the percent above 75%
 - Non-numeric  Frequency values
 - Numeric  Median



DATA PREPROCESSING

Data Standardization

“Rating”
column



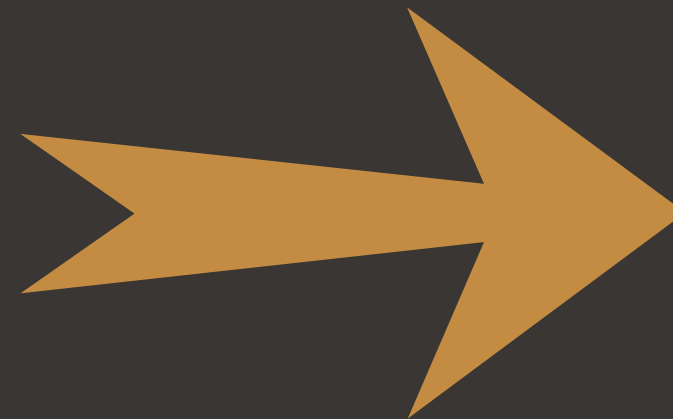
NULL
values

“unknown”
values



DATA PREPROCESSING

Data Standardization



DATA EXPLORATION & PREPROCESSING

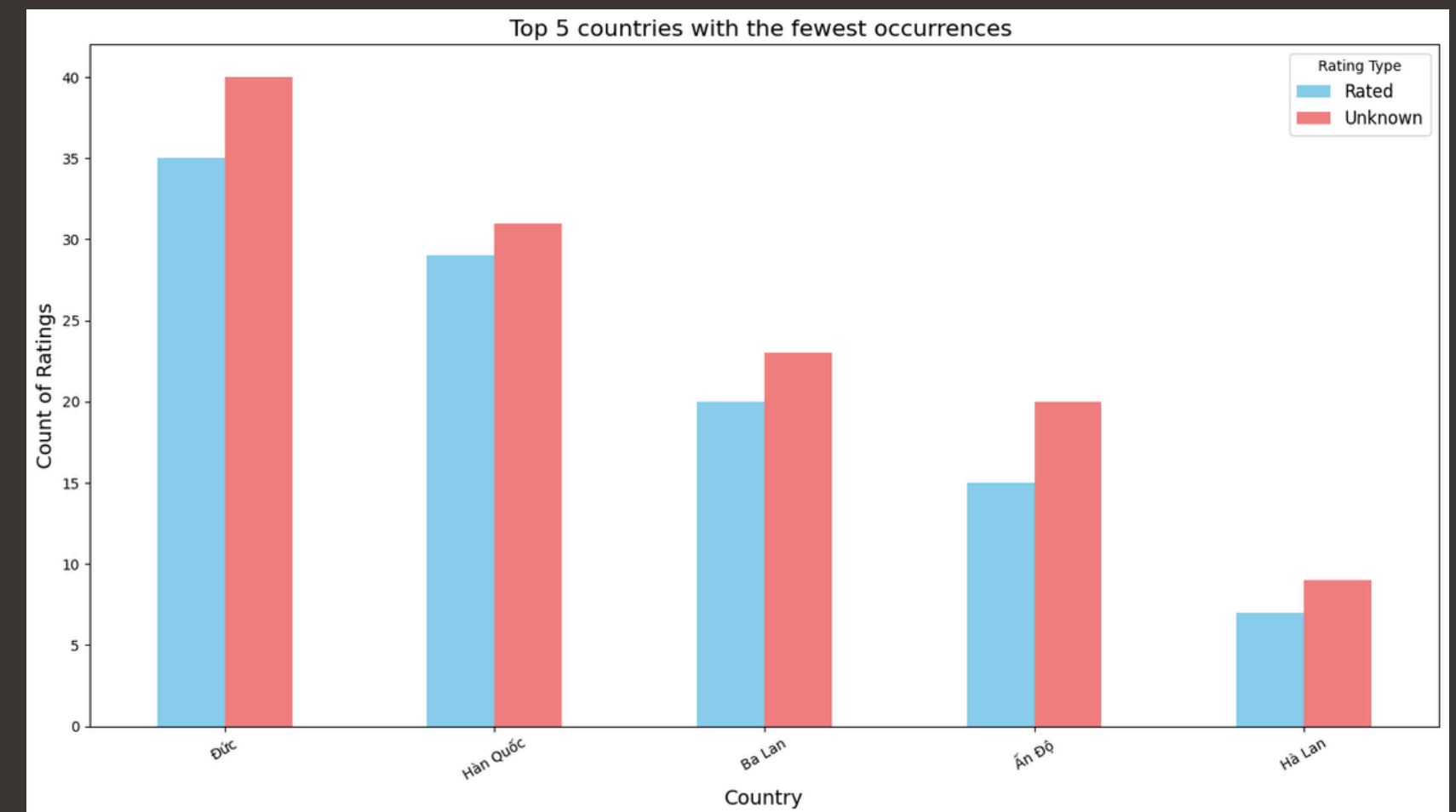
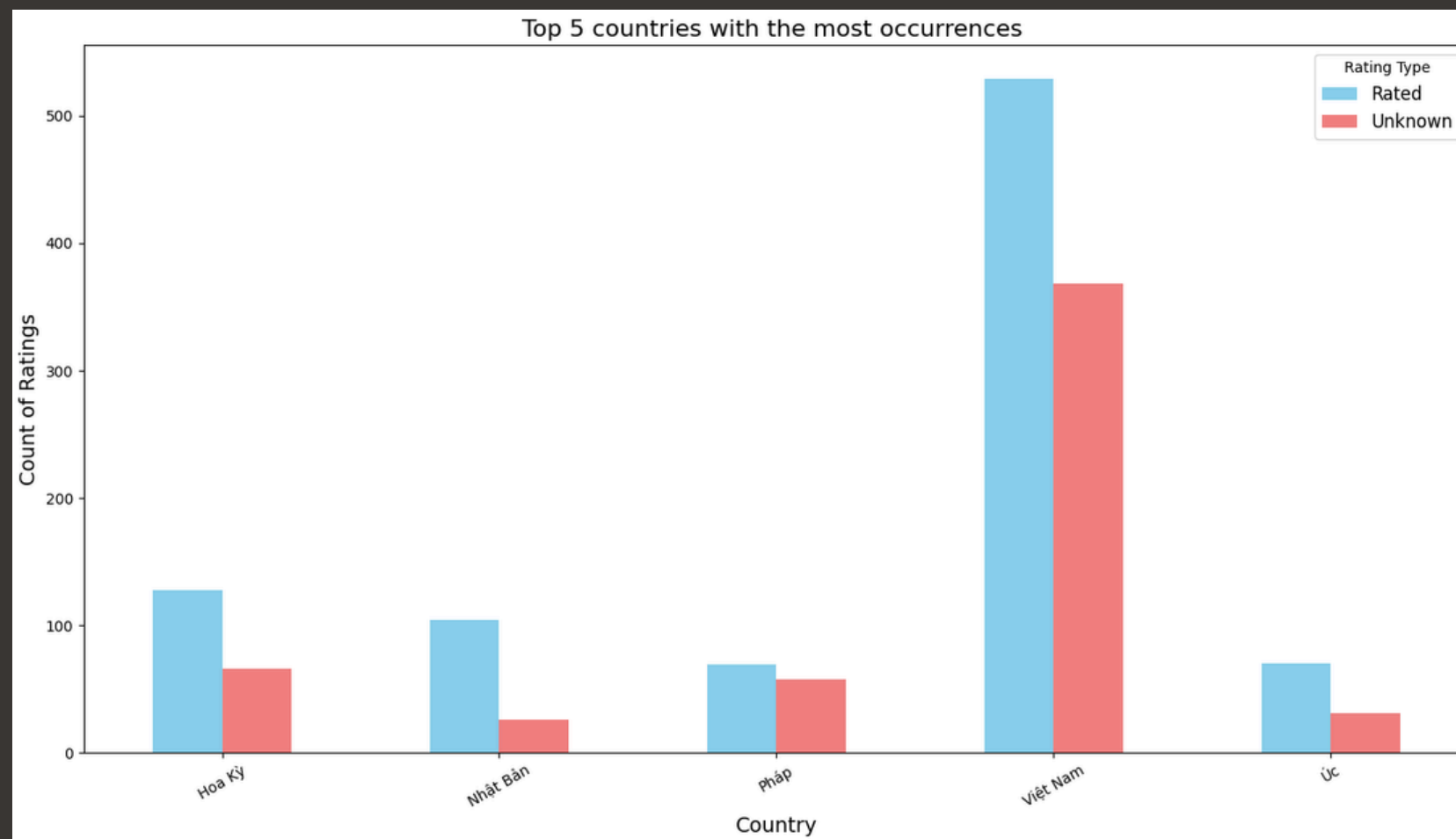
Questions

- 1 In each category, which factor is decisive?
- 2 In each category, which domestic and foreign products receive more attention?
- 3 In each category, the distribution of prices.
- 4 In each category, the distribution of brands.
- 5 In each category, the distribution of dosage forms.

DATA EXPLORATION & PREPROCESSING



Question 1: In each category, which factor is decisive?

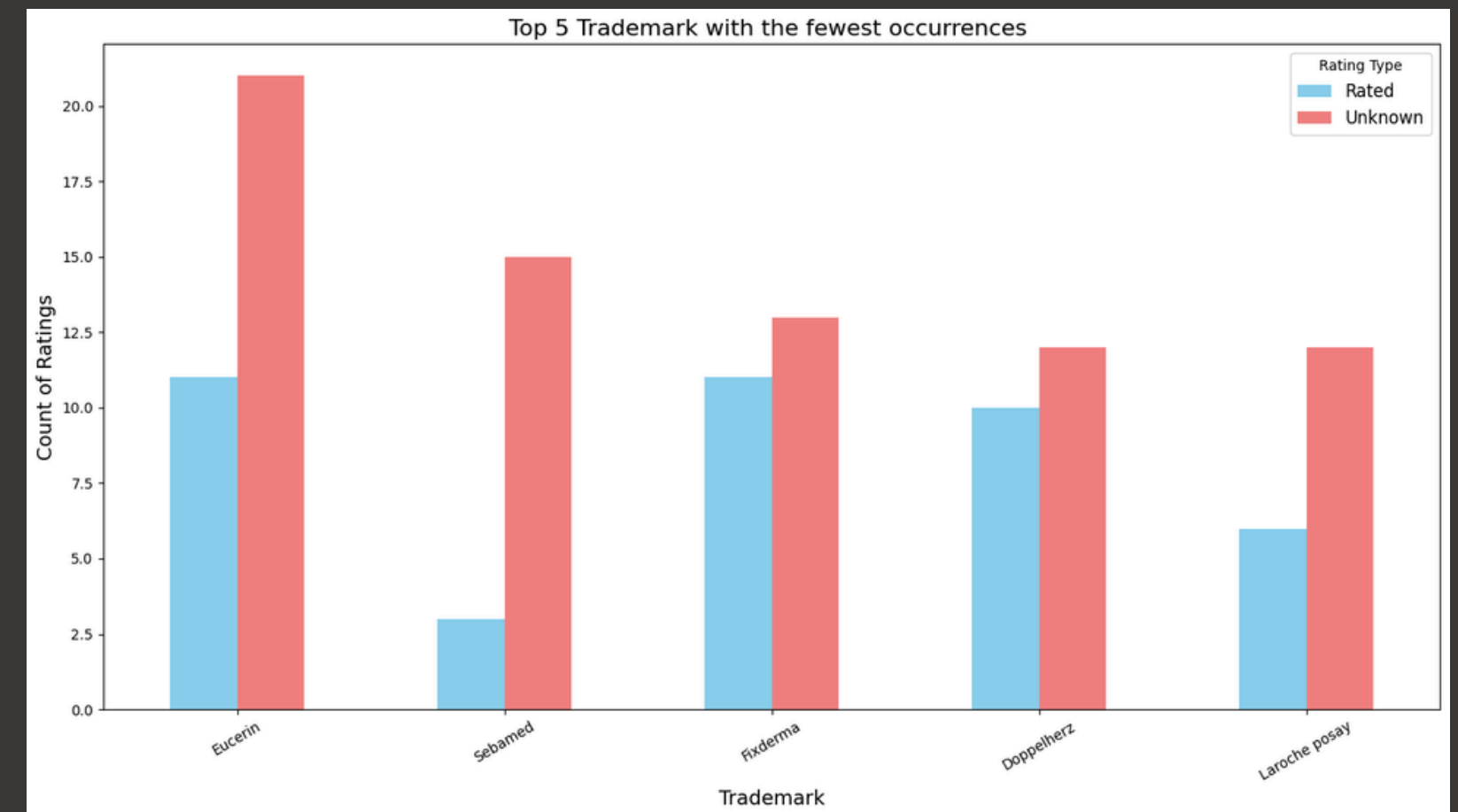
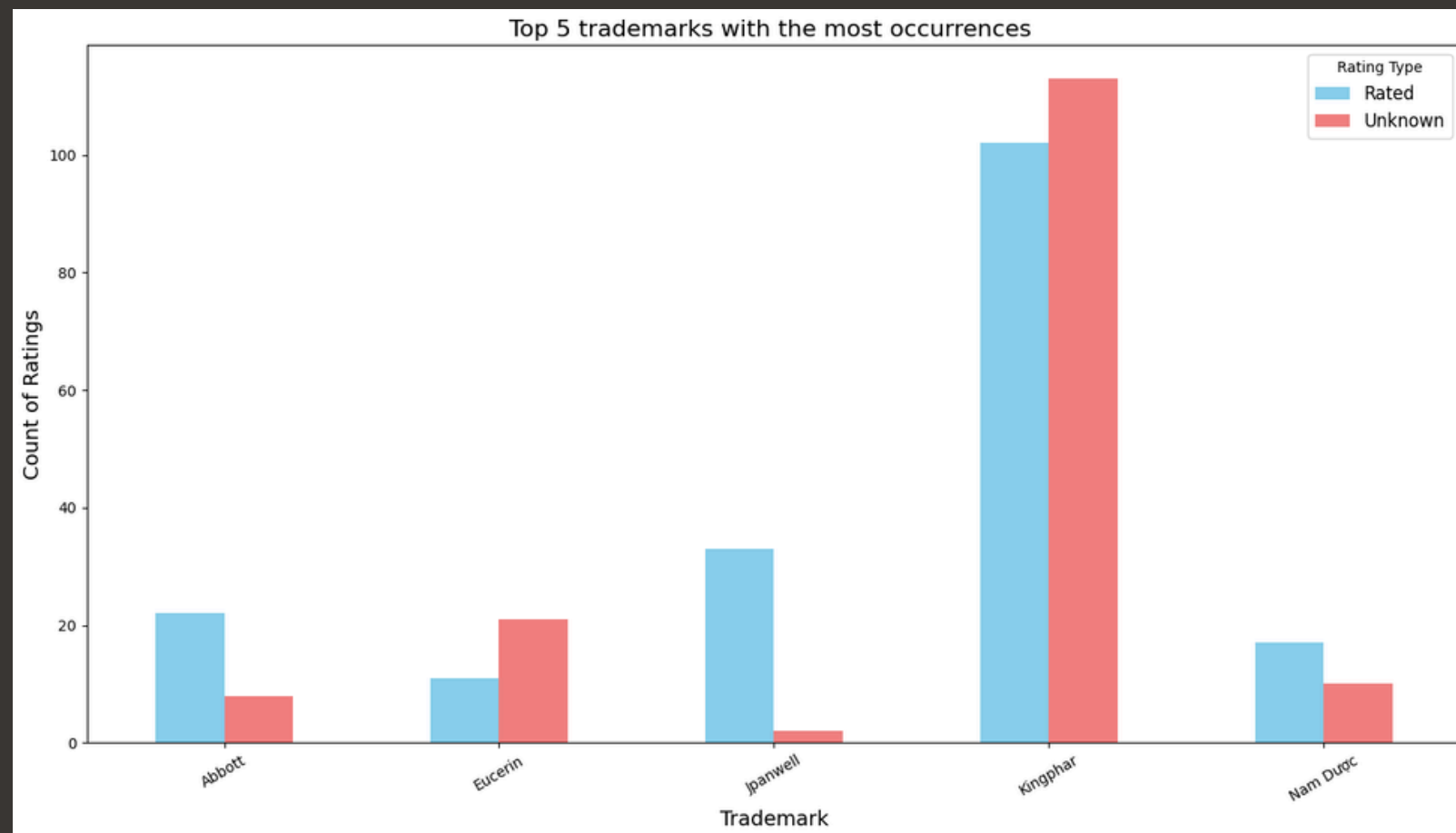


Country

DATA EXPLORATION & PREPROCESSING



Question 1: In each category, which factor is decisive?

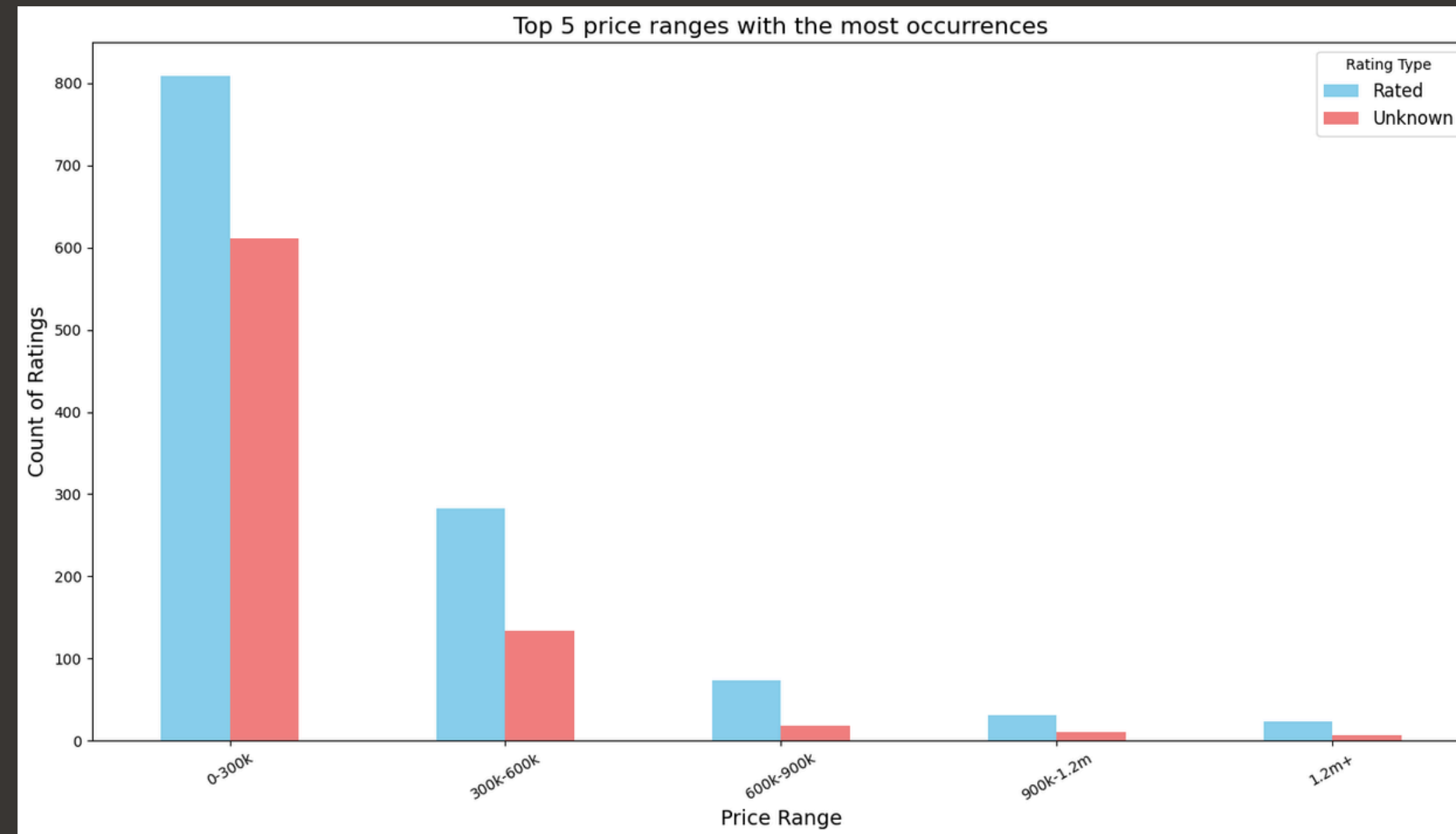


Trademark

DATA EXPLORATION & PREPROCESSING



Question 1: In each category, which factor is decisive?

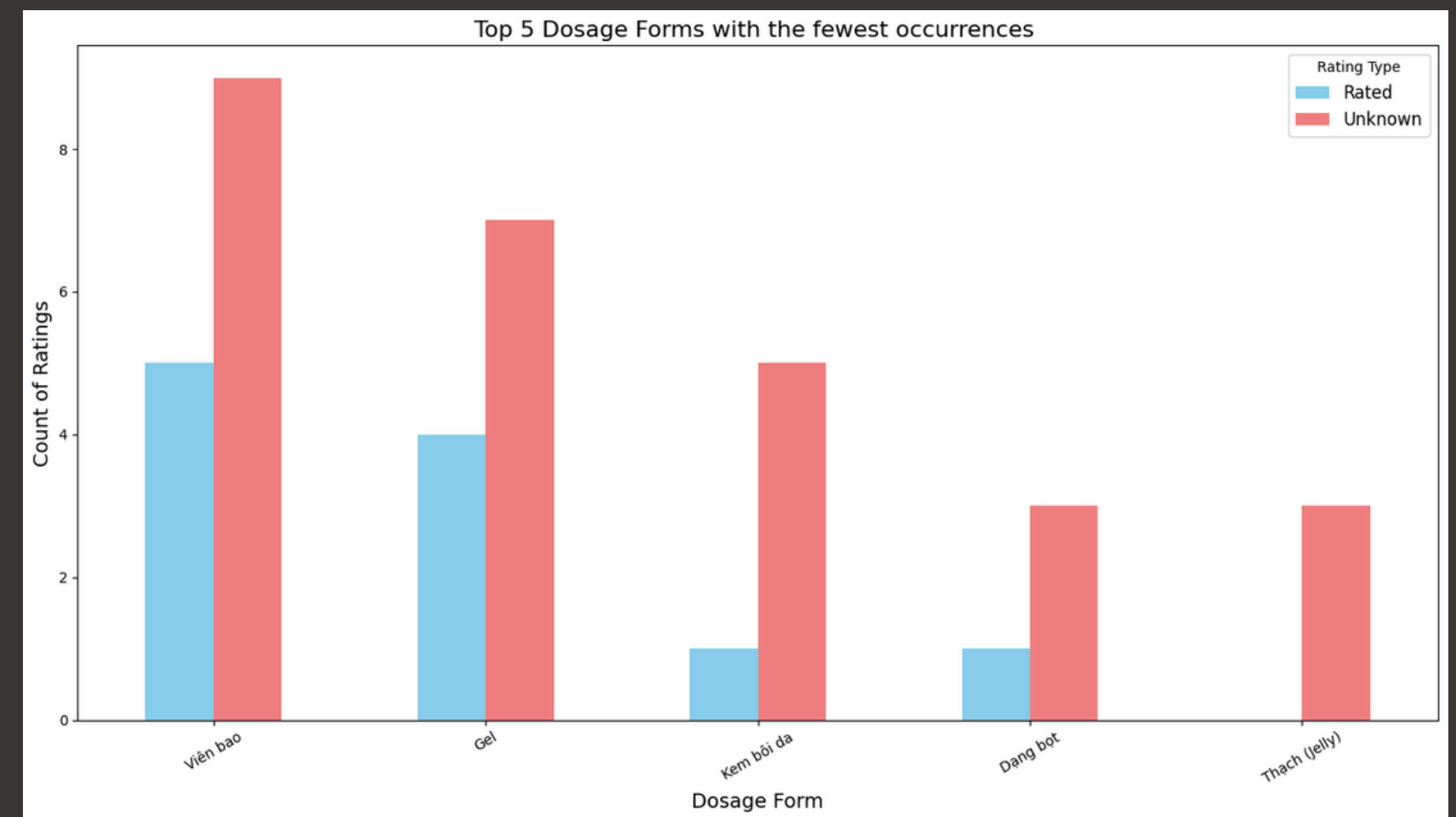
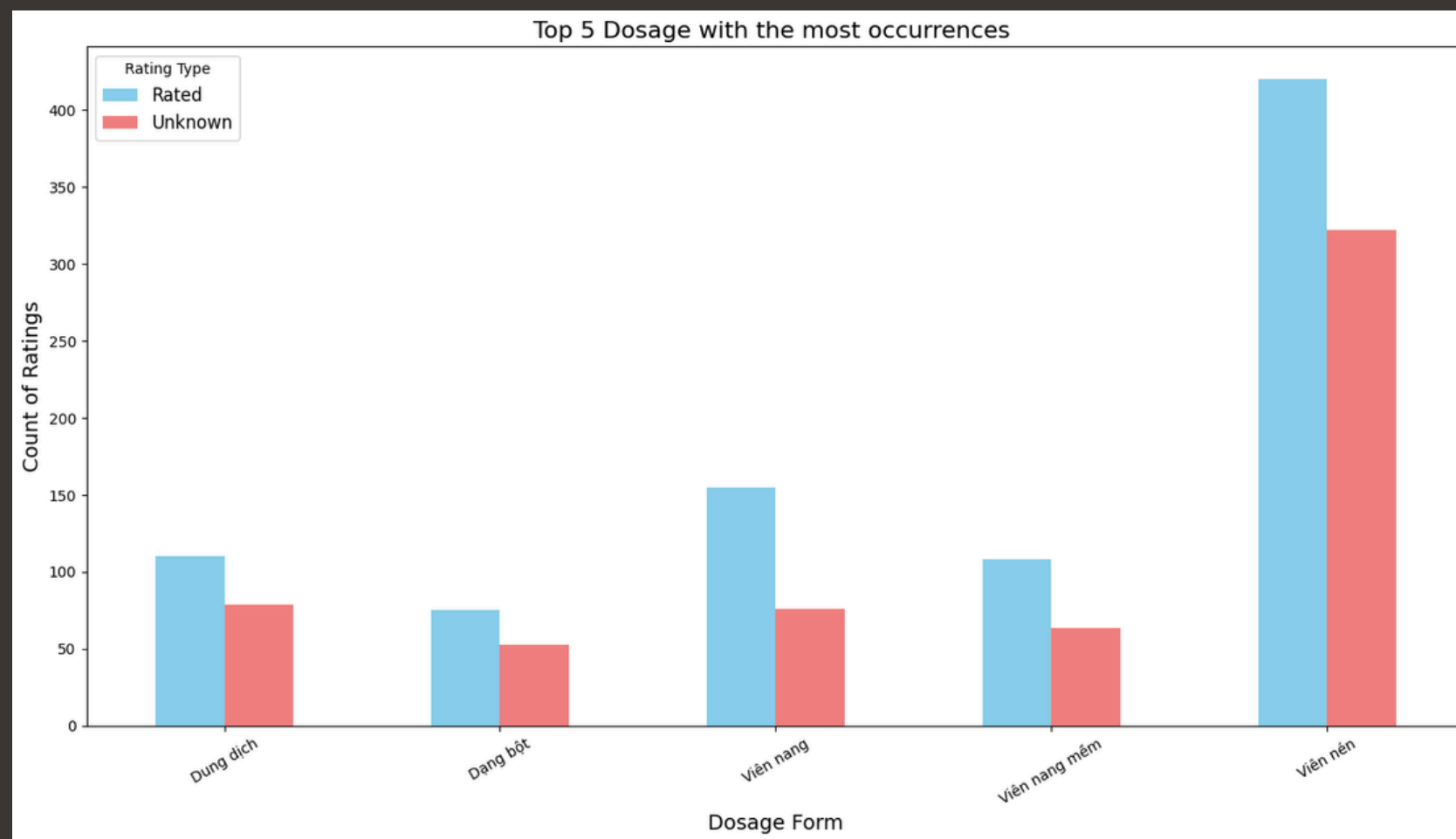


Price

DATA EXPLORATION & PREPROCESSING



Question 1: In each category, which factor is decisive?

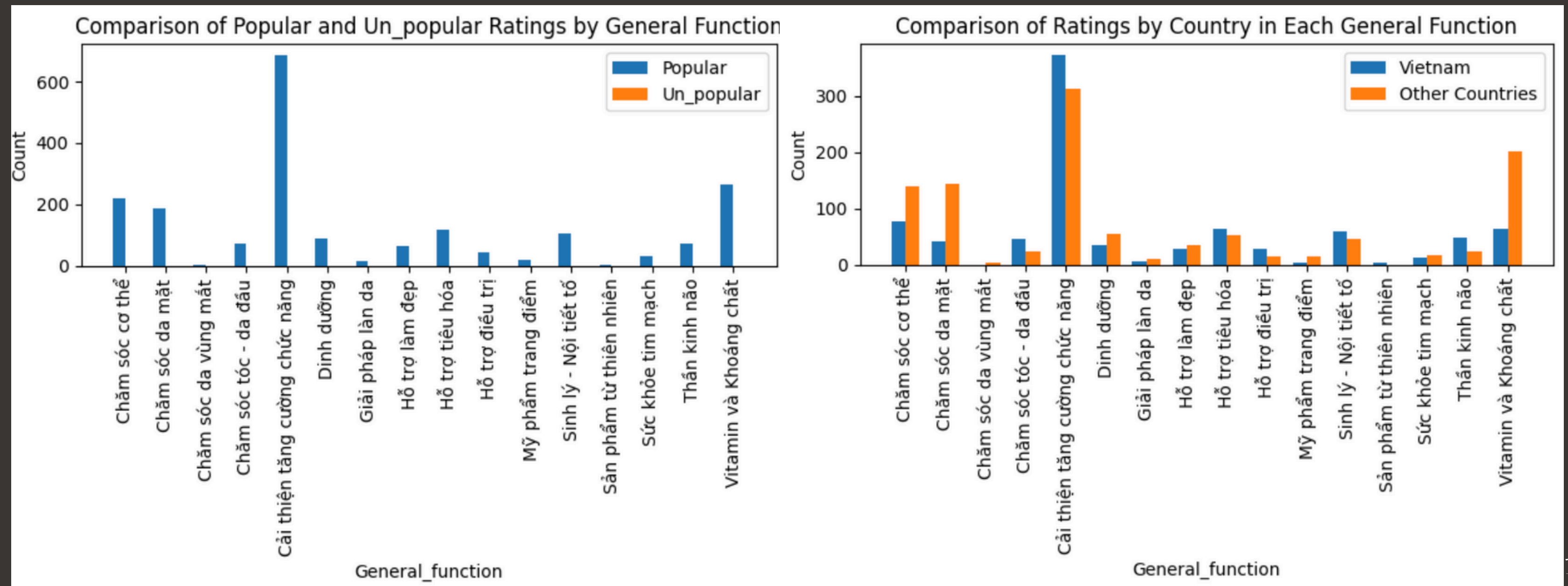


Dosage Form

DATA EXPLORATION & PREPROCESSING



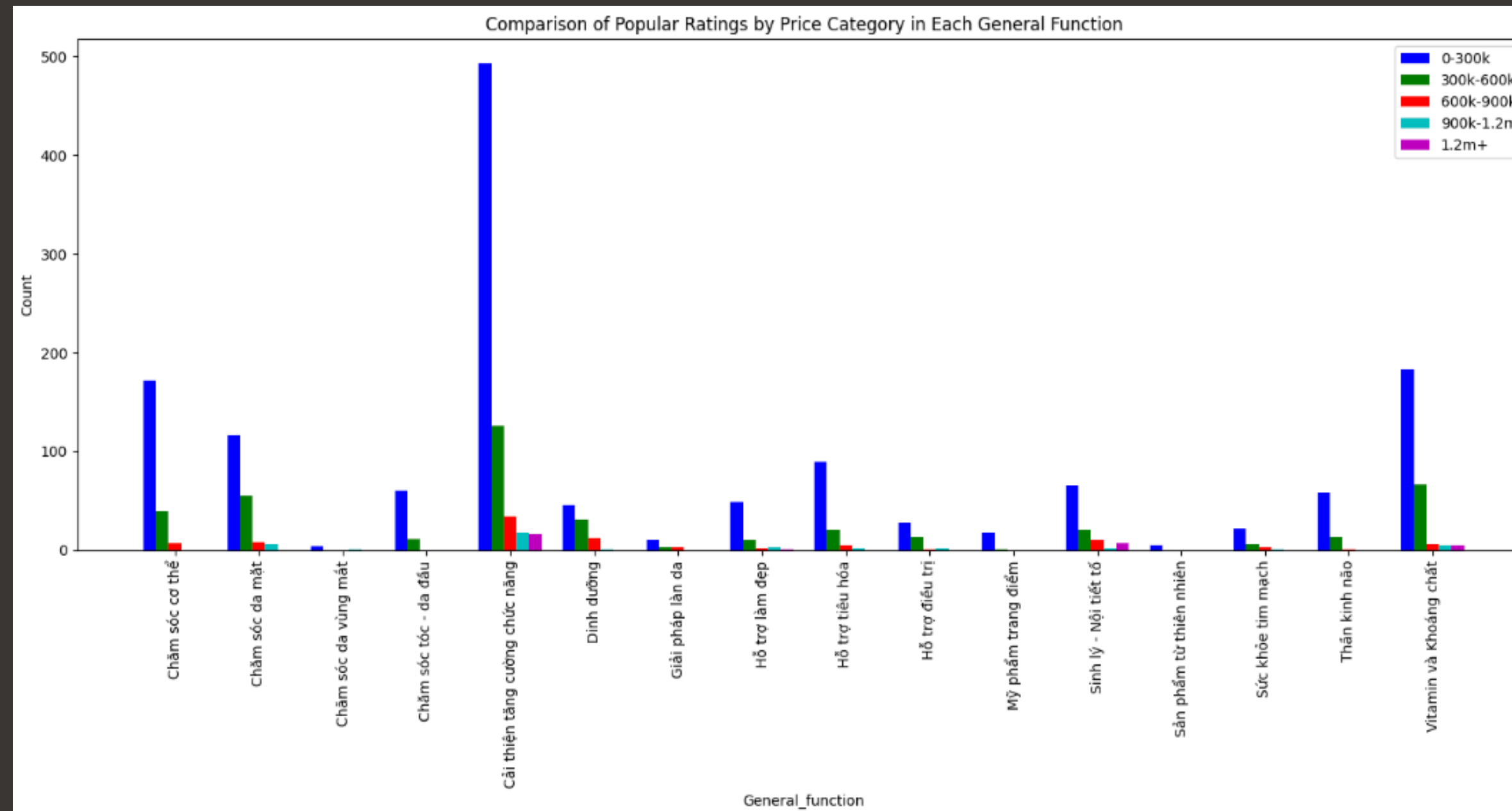
Question 2 In each category, which domestic and foreign products receive more attention?



DATA EXPLORATION & PREPROCESSING



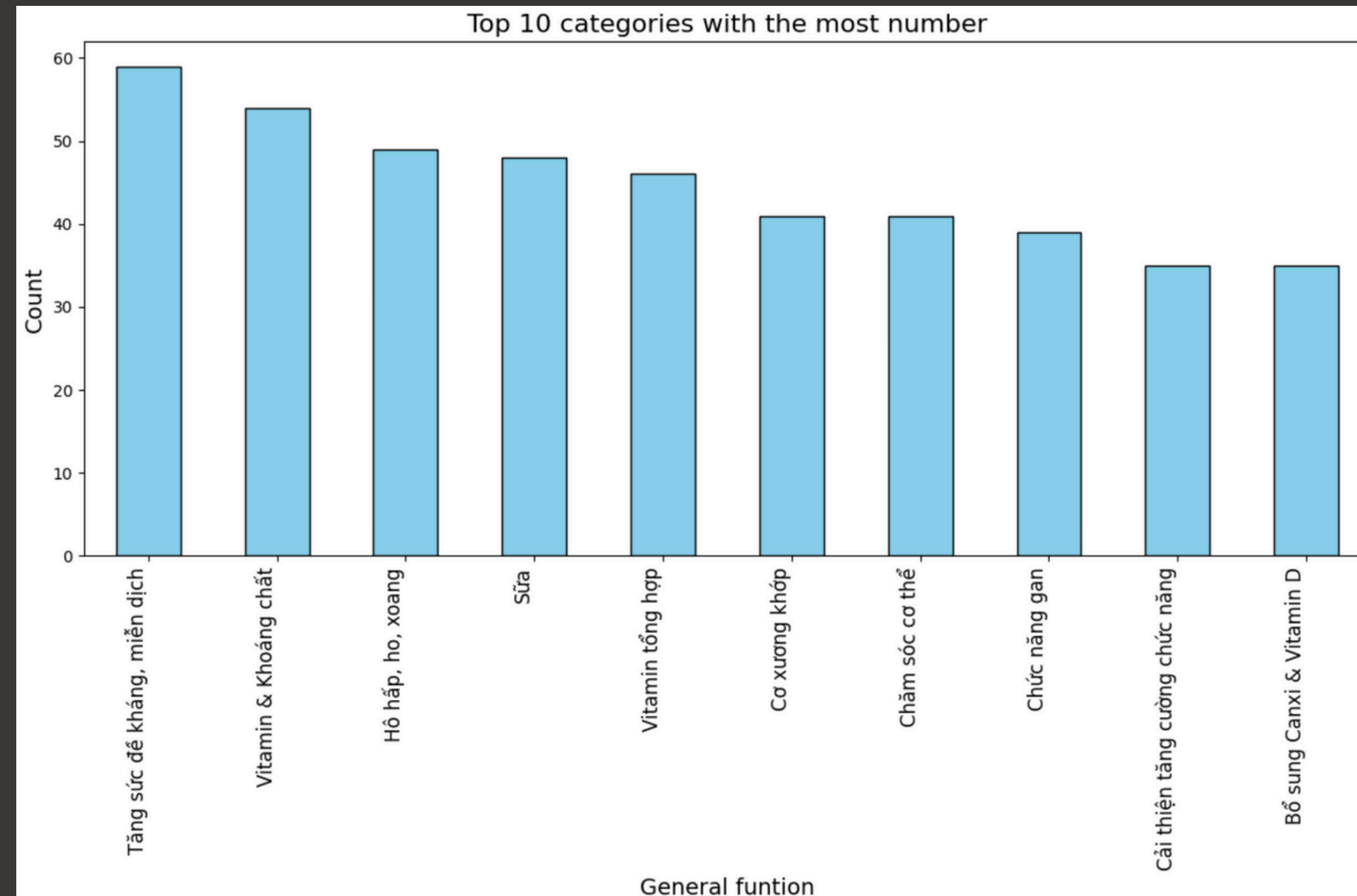
Question 3 In each category, the distribution of prices.



DATA EXPLORATION & PREPROCESSING



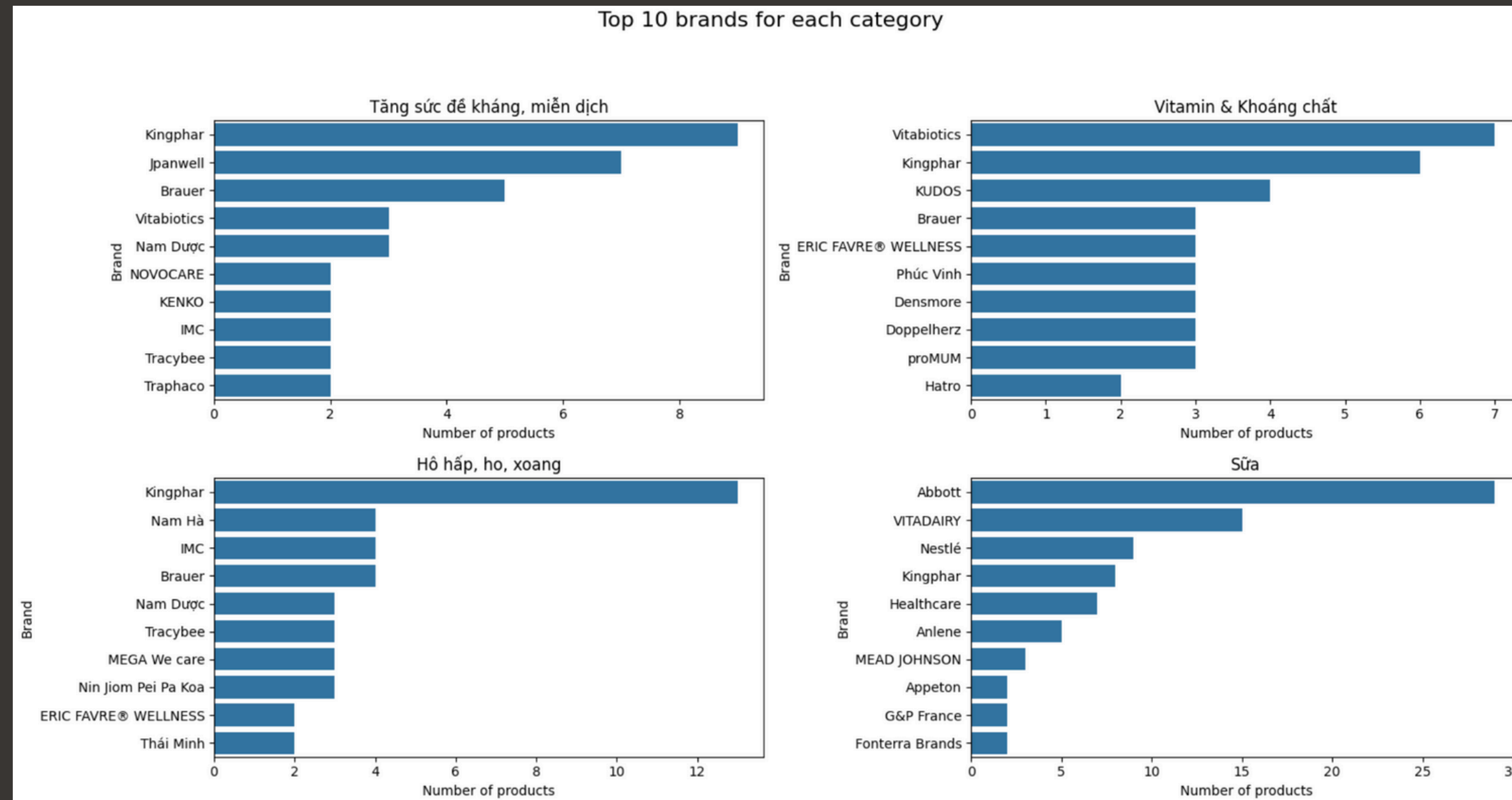
Question 4 In each category, the distribution of brands.



DATA EXPLORATION & PREPROCESSING



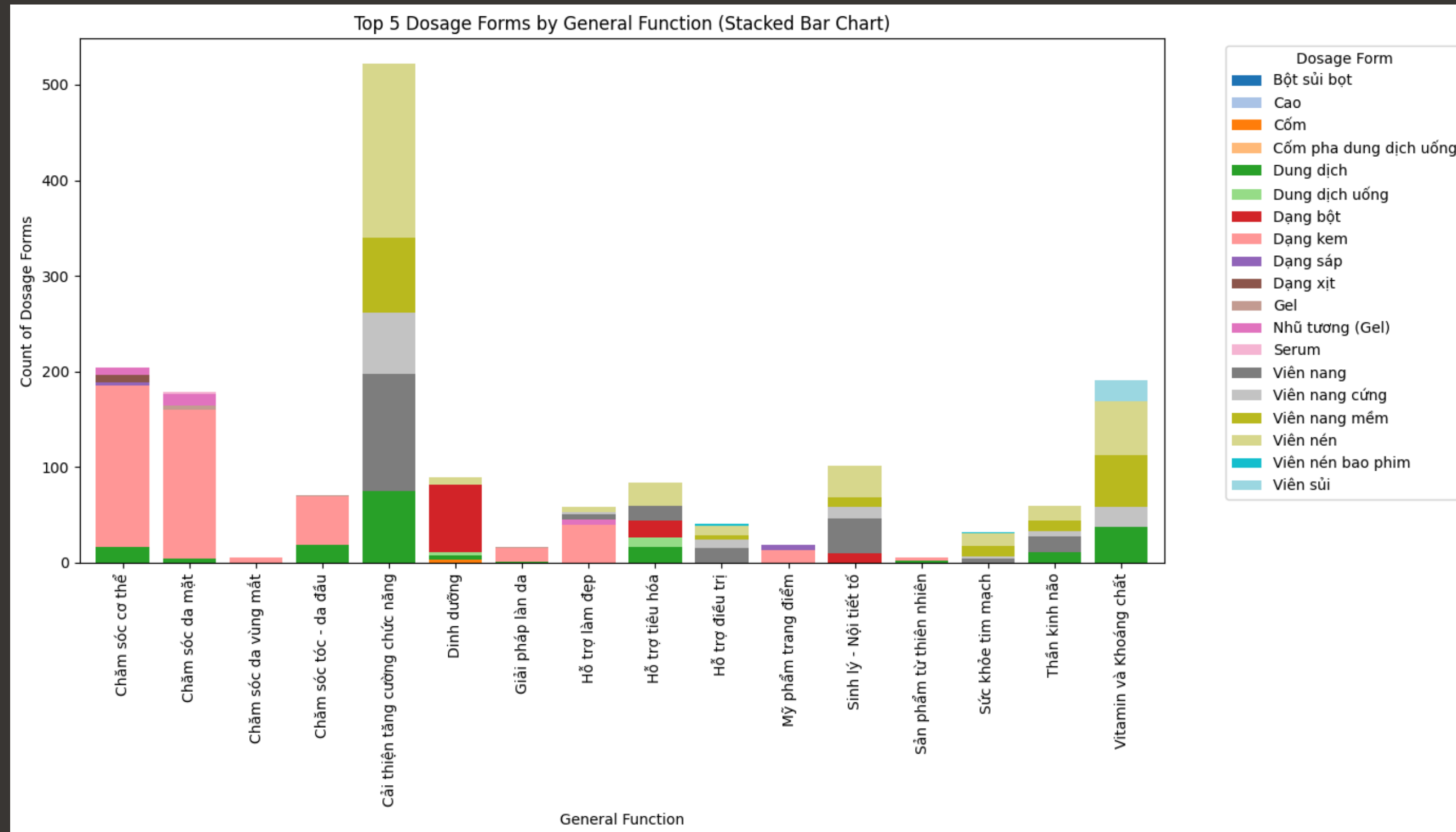
Question 4 In each category, the distribution of brands.



DATA EXPLORATION & PREPROCESSING



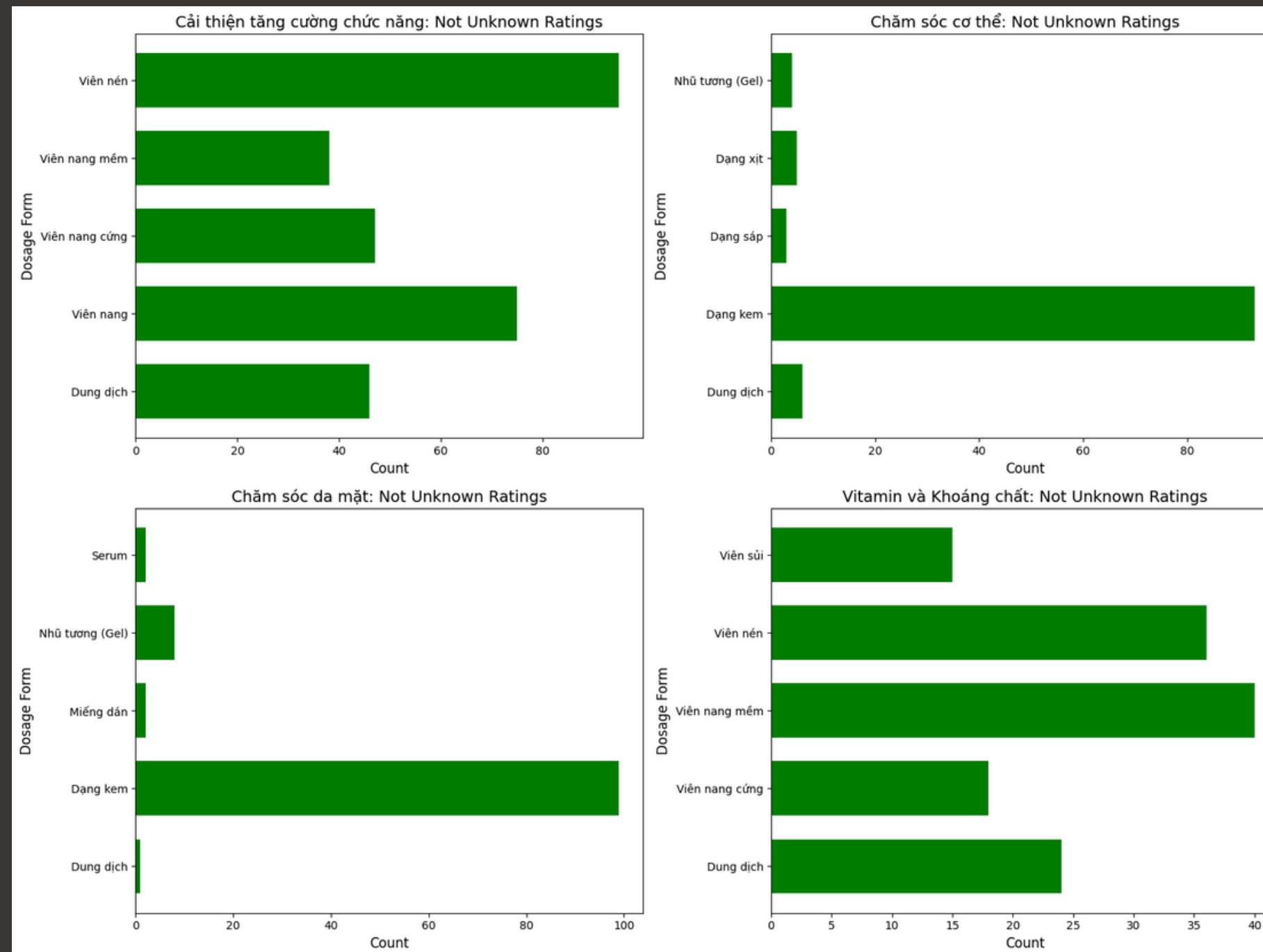
Question 5 In each category, the distribution of dosage forms.



DATA EXPLORATION & PREPROCESSING



Question 5 In each category, the distribution of dosage forms.





MODEL BUILDING

Ways To Define Models





MODEL BUILDING

Build Models

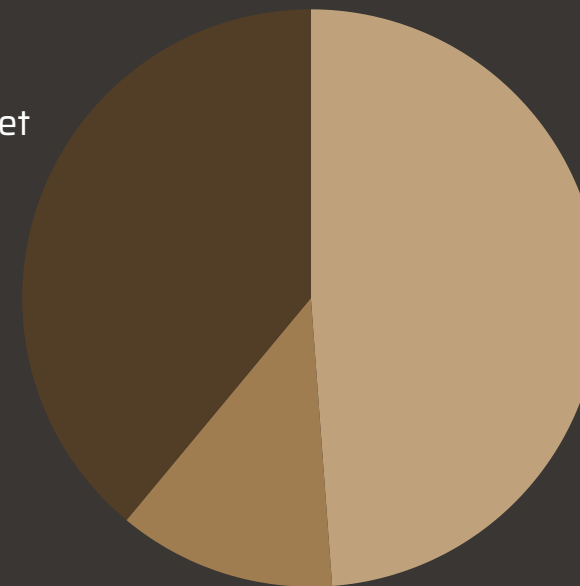


Choose features
may affect
customer demands

Tree		Type
		1
		2
		1
		2
		3

Encodes categorical
labels into integer
values.

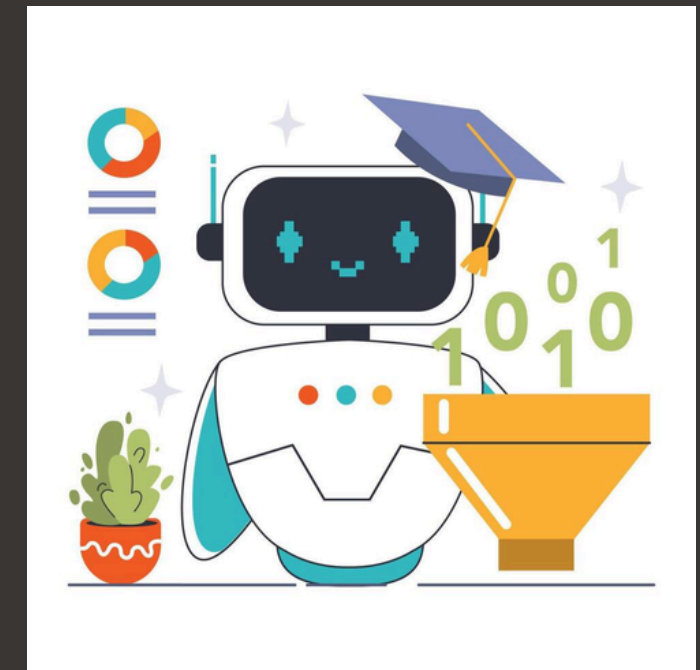
Prediction dataset
39%



Test dataset
12.2%

Train dataset
48.8%

Split data



Train models



FINAL RESULTS

Model Comparisions - RandomizedSearchCV

Model	RMSE	MSE	TrainingTime	Memory Usage (MB)
Ridge	0.36139	0.13060	1.9744	137.06
ElasticNet	0.36145	0.13065	1.9912	137.28
SVR	0.36162	0.13077	1.9796	136.17
RandomForestRegressor	0.36776	0.13524	1.6031	131.00
GradientBoostingRegressor	0.37403	0.13989	1.3402	132.65



FINAL RESULTS

Use Best Model To Predict Unrating Data



Ridge Regression Model

Price	Trademark	Country	General_function	Predicted Rating
390000.0	375	23	0	4.947837
22400.0	214	38	3	4.920278
151200.0	290	38	0	4.921739
39000.0	14	38	0	4.900257
39000.0	14	38	0	4.900257
...	...	...	...	...
276000.0	208	38	14	4.861462
276000.0	342	38	4	4.885514
276000.0	342	38	15	4.934686
276000.0	234	7	5	4.946959
276000.0	157	8	15	4.873883

**THANK YOU FOR YOUR
ATTENTION**

